

OPTIMIZING HEALTHCARE RESOURCE ALLOCATION FOR DIABETIC PATIENTS IN KENYA USING MACHINE LEARNING

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INTRODUCTION

This project uses machine learning to identify high-risk diabetic patients in Kenya, enabling targeted interventions, efficient resource allocation, and improved patient outcomes..



PROBLEM AND ITS IMPACT

Kenya's public healthcare system faces high diabetic readmissions due to poor early risk identification and limited resources.

This project enables data-driven prioritization of high-risk patients, improving resource efficiency by **40–60%**, reducing readmissions by **20–30%**, lowering healthcare costs by **15–25%**, easing hospital congestion, and supporting informed national and county healthcare planning.

PROJECT OBJECTIVES



Develop a machine learning model



Identify diabetic patients at high risk of 30-day readmission before discharge



Enable targeted clinical interventions



Improve efficiency in healthcare resource allocation



Enhance patient outcomes within the healthcare system

DATA UNDERSTANDING AND CLEANING



DATA
UNDERSTANDING



TARGET VARIABLE
PREPARATION



ID MAPPING AND
DATA CLEANING



MISSING VALUE
HANDLING



DATA QUALITY
VERIFICATION

DATA UNDERSTANDING

The dataset used in this project contains **101,766 patient encounters** related to diabetic hospital admissions, originally collected from **130 hospitals**. Each record represents a single hospital encounter and includes **demographic, clinical, and utilization-related features** relevant to predicting 30-day readmission.

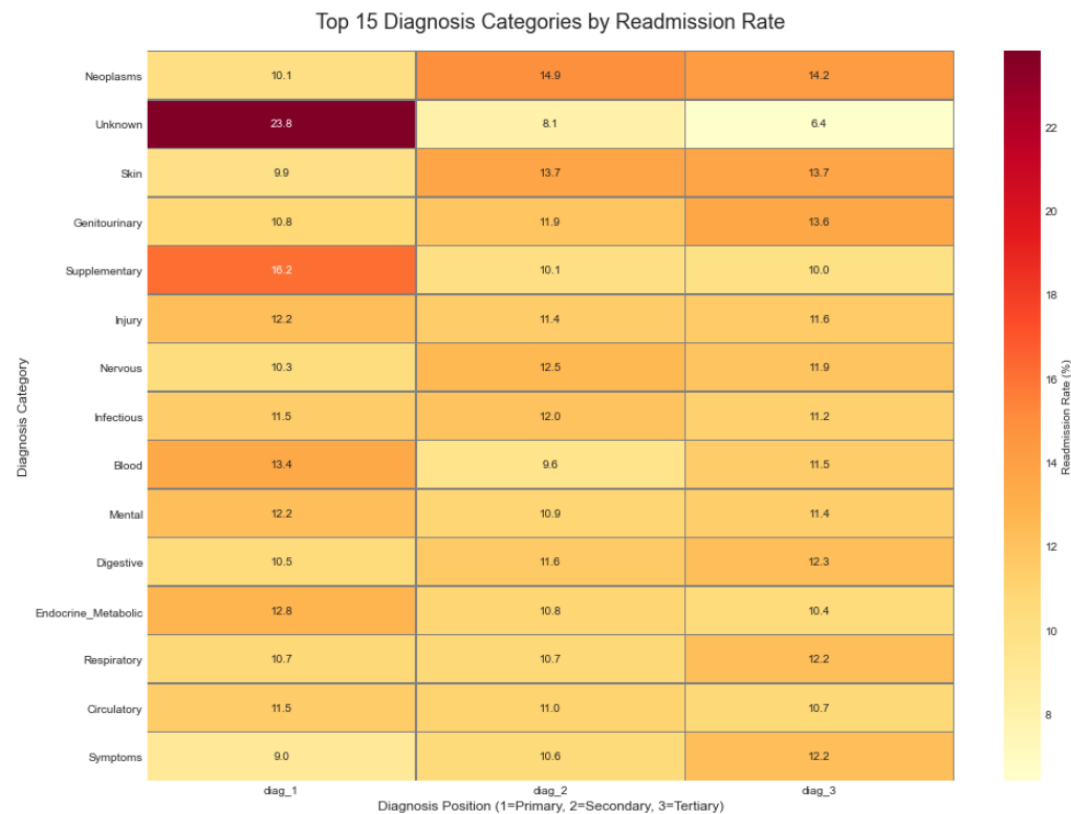
Features: 50 variables

Target Variable: 30-day hospital readmission (binary classification)

Data Type Mix:

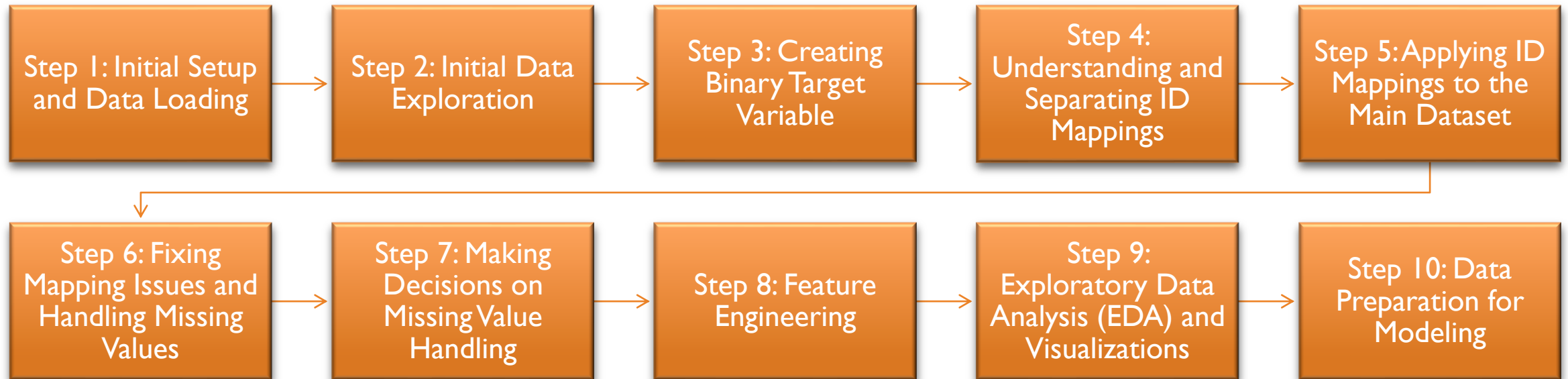
- Categorical (race, gender, admission type, discharge disposition)
- Numerical (time in hospital, number of medications, lab procedures)

TOP 15 DIAGNOSIS CATEGORIES BY READMISSION RATE



- Readmission rates vary across diagnosis categories and diagnosis positions, with higher risk observed in more complex or unclear conditions. This highlights the importance of diagnosis-related features in identifying patients at higher risk of readmission.

DATA CLEANING (CRISP-DM) METHODOLOGY



MODELLING

TRAINED THREE MACHINE LEARNING MODELS TO PREDICT 30-DAY DIABETIC READMISSION

MODELS USED: LOGISTIC REGRESSION (BASELINE & SMOTE), RANDOM FOREST, AND XGBOOST

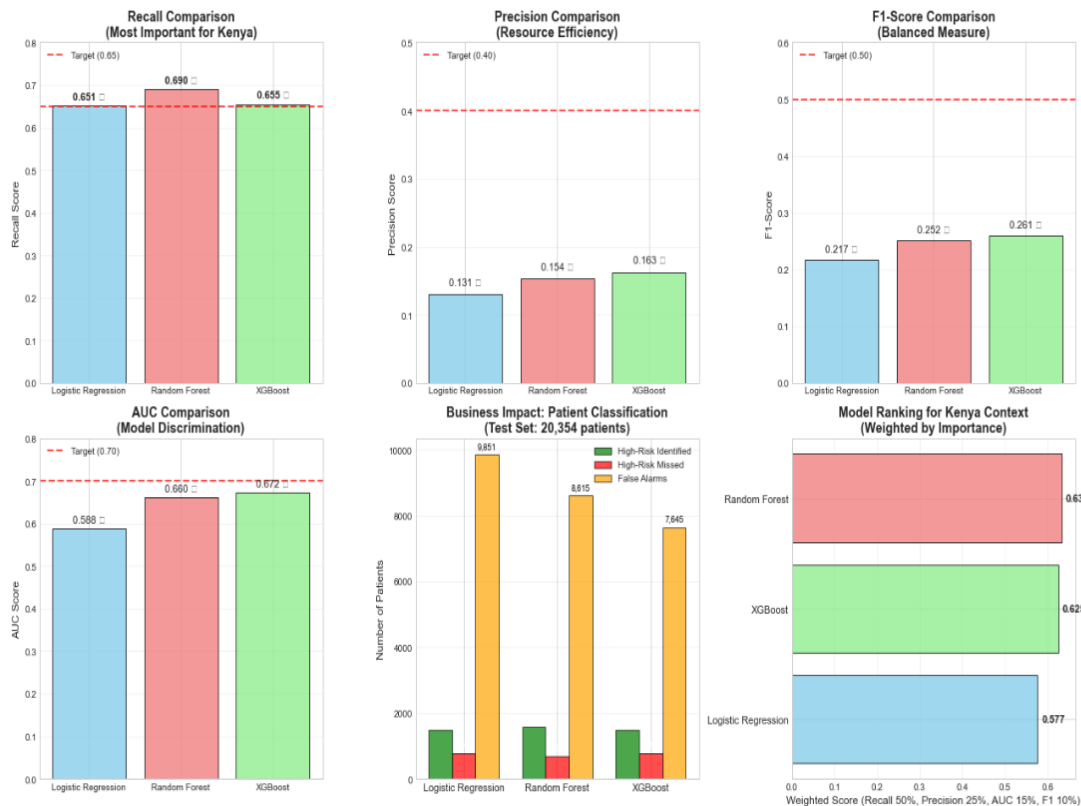
ADDRESSED CLASS IMBALANCE IN THE DATASET

PERFORMED HYPERPARAMETER TUNING FOR MODEL OPTIMIZATION

OPTIMIZED CLASSIFICATION THRESHOLD TO MAXIMIZE RECALL AND IDENTIFY HIGH-RISK PATIENTS

MODELS AND COMPARISON

Model Comparison: Logistic Regression vs Random Forest vs XGBoost



- **Logistic Regression:** Baseline model with acceptable recall but low precision and overall accuracy; useful for interpretability.
- **Random Forest:** Best overall performer with the **highest recall**, identifying the most high-risk patients and minimizing missed cases.
- **XGBoost:** Most balanced model with the **highest precision, F1-score, and AUC**, reducing unnecessary interventions.
- **Key Takeaway**
- Random Forest is the **preferred model** for Kenya's healthcare context due to its strong recall and clinical impact, while XGBoost offers a strong efficiency-focused alternative.

PERFORMANCE METRICS

- Random Forest achieves the **highest recall**, making it most effective at identifying high-risk patients. XGBoost provides the **best precision and AUC**, offering better overall discrimination, while Logistic Regression serves as a **baseline model** with lower predictive performance.

Performance Metrics

Model	Recall	Precision	AUC
Logistic Regression	0.651	0.131	0.588
Random Forest	0.690	0.154	0.660
XGBoost	0.655	0.163	0.672

MODEL SELECTION

- In this project, **recall** was prioritized over precision because the primary goal is to **identify as many high-risk diabetic patients as possible before discharge**. Missing a high-risk patient may lead to preventable complications, readmissions, and increased healthcare costs. In contrast, false positives only result in additional monitoring or follow-up, which is acceptable in a resource-constrained healthcare setting. Therefore, maximizing recall ensures patient safety and supports proactive, targeted interventions.

- **Recommended Model: Random Forest**

- **Rationale:**

Achieved the **highest recall (69%)**, maximizing identification of high-risk patients

Maintained **acceptable precision**, ensuring efficient use of limited resources

Demonstrated **strong generalization** across evaluation metrics

Offers better **interpretability and deploy ability** compared to more complex models

Random Forest provides the best balance between **patient safety, operational feasibility**, and **policy relevance** within Kenya's healthcare context.

DEPLOYMENT

OBJECTIVE

- To operationalize the trained **Random Forest model** as a **decision-support tool** that identifies diabetic patients at high risk of 30-day readmission before hospital discharge.
- **Deployment Platform**
- Web-based application built using **Streamlit**
- Accessible via browser:
<https://kenya-hospital-streamlit.onrender.com/>
- Designed for use by healthcare workers in real-time clinical settings

HOW THE DEPLOYMENT WORKS

Learning Framework

- Trained on **81,412 patient cases**
- Identifies patterns associated with 30-day readmission risk
- Applies **weighted analysis**, recognizing that some factors have greater influence than others
- Continuously improves as new patient data becomes available

Key Predictors of Readmission Risk

1. Prior Hospital Visits (48% influence)

Higher utilization strongly increases readmission risk

2. Discharge Destination (14–15% influence)

Post-discharge care setting significantly affects outcomes

3. Emergency Department Visits (3.5% influence)

Recent ER visits indicate clinical instability

Model Output & Decision Logic

- **Risk Score:** Generates a **0–100% probability** of 30-day readmission
- **Decision Threshold: 48%**, optimized for Kenya's healthcare context
- **Final Output:** Classifies patients as **High-Risk** or **Low-Risk** to guide intervention planning

CLINICAL & OPERATIONAL USE

Decision Support System

Risk-Based Care Pathways

High-Risk Patients (Risk > 48%)

- Priority follow-up within **7 days**
- Coordinated multidisciplinary care
- Home-based care services arranged where appropriate

Low-Risk Patients (Risk ≤ 48%)

- Standard **30-day** follow-up
- Routine discharge planning
- Patient education and self-management guidance

Illustrative Patient Scenarios

■ Patient A

- Multiple hospital visits in the past year
- Polypharmacy and complex discharge needs

■ Predicted Risk: 78%

- **Action:** Intensive post-discharge intervention

■ Patient B

- First hospital admission
- Short length of stay
- Discharged home

■ Predicted Risk: 25%

- **Action:** Standard discharge and follow-up care

Model Performance & Relevance

- **69% recall**, successfully identifying most high-risk patients
- Designed to **prioritize sensitivity**, minimizing missed critical cases
- **Optimized for Kenya's healthcare context**, ensuring practical and policy-relevant deployment

OPERATIONAL BENEFITS

IMPLEMENTATION IMPACT

Resource Optimization

- Efficient targeting of limited clinical staff time
- Reduction of unnecessary follow-ups
- Focus on patients with the highest care needs

Quality Improvement

- Standardized, data-driven risk assessment
- Reduction in preventable readmissions
- More consistent clinical decision-making

Patient Outcomes

- Earlier intervention for high-risk patients
- Improved continuity of post-discharge care
- Better overall health outcomes

CONCLUSION

- Successfully developed a machine learning model to predict 30-day readmission among diabetic patients
- Identified key clinical and utilization factors influencing readmission risk
- Random Forest achieved the **highest recall**, meeting project objectives
- Demonstrated that data-driven risk stratification can improve resource allocation in Kenya's healthcare system

RECOMMENDATIONS

- Deploy the Random Forest model as a **decision-support tool** at discharge
- Use predictions to prioritize high-risk patients for targeted interventions
- Integrate the model with hospital information systems for routine use
- Monitor model performance and retrain periodically with local data
- Extend the approach to other chronic diseases

THANK YOU

G4: FAITH, GODFREY ,DIANA , CINDY, EDNAH, MITCHELLE, ELSIE.

